

LAB EXPERIMENT RECORD

Set 5: Website Analytics

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	DIVYASREE.M
Register Number	192524098
Date	1 October 2026
Day	Thursday

Aim

To analyse website traffic data using a line chart, bar chart and stacked area chart, and to build an interactive dashboard combining the line and bar charts.

Dataset: Website Traffic

Date	Page Views	Click-through Rate (%)
2023-01-01	1500	2.3
2023-01-02	1600	2.7
2023-01-03	1400	2.0
2023-01-04	1650	2.4
2023-01-05	1800	2.6

Procedure

1. Create the date, page views and click-through rate vectors, converting dates with `as.Date()`.
2. Use `plot()` to draw the daily page views line chart.
3. Use `order()` to find the top N days by click-through rate and `barplot()` to display them.
4. Create likes, shares and comments vectors, compute cumulative layers and draw the stacked area chart with `polygon()`.
5. Build a Shiny dashboard with a date-range filter and a Top N input showing the line and bar charts together, and run it with `shinyApp()`.

Key R Functions Used

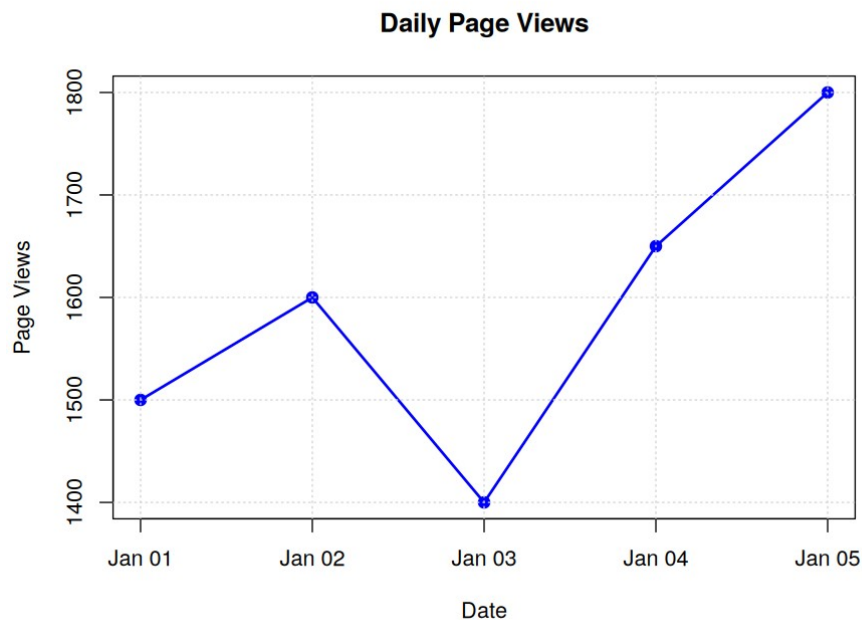
Function	Purpose
<code>as.Date()</code>	Converts text values to dates.
<code>plot()</code> / <code>grid()</code>	Create the line chart and plot area.
<code>order()</code>	Sorts values to find the top N days.
<code>barplot()</code> / <code>text()</code>	Create the bar chart and value labels.
<code>polygon()</code> / <code>legend()</code>	Fill the stacked area layers and add a legend.
<code>dateRangeInput()</code> / <code>renderPlot()</code> / <code>shinyApp()</code>	Build and run the dashboard.

PROGRAM 1: Line Chart of Daily Page Views

Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
# Dataset
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                  "2023-01-04", "2023-01-05"))
page_views <- c(1500, 1600, 1400, 1650, 1800)
click_rate <- c(2.3, 2.7, 2.0, 2.4, 2.6)

# Line Chart
plot(date, page_views,
     type = "o",
     pch = 19,
     lwd = 2,
     col = "blue",
     xlab = "Date",
     ylab = "Page Views",
     main = "Daily Page Views")
grid()
```



Output 1: Daily Page Views (Line Chart)

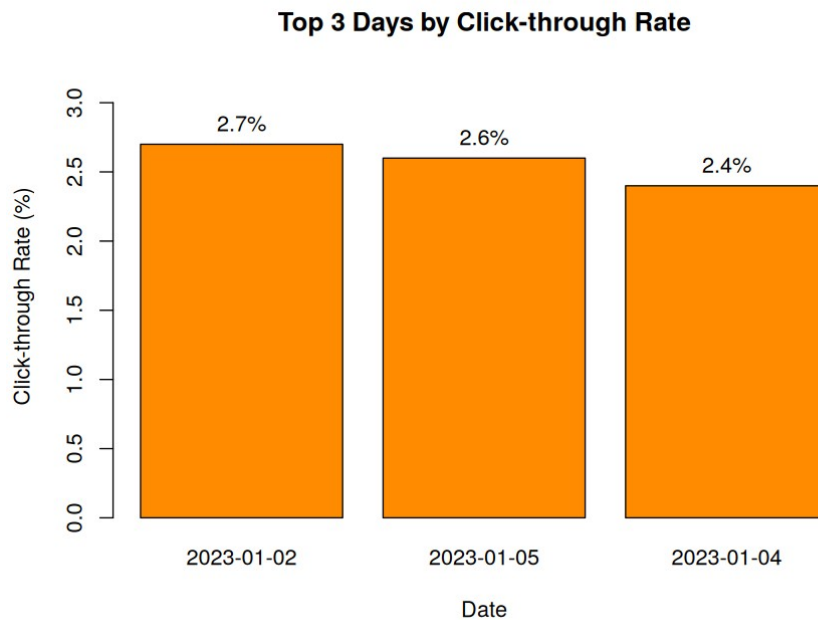
Observation: Page views rise to 1,600 on 2 January, drop to the lowest value of 1,400 on 3 January, then climb steadily to the highest value of 1,800 on 5 January. Overall traffic is increasing.

PROGRAM 2: Bar Chart of Top N Days by Click-through Rate

Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
# Top N days (N = 3) by click-through rate
N <- 3
top_days <- order(click_rate, decreasing = TRUE)[1:N]

# Bar Chart
bp <- barplot(click_rate[top_days],
              names.arg = format(date[top_days]),
              col = "darkorange",
              ylim = c(0, 3.2),
              xlab = "Date",
              ylab = "Click-through Rate (%)",
              main = paste("Top", N, "Days by Click-through Rate"))
text(bp, click_rate[top_days],
     labels = paste0(click_rate[top_days], "%"), pos = 3)
```



Output 2: Top 3 Days by Click-through Rate (Bar Chart)

Observation: 2 January has the highest click-through rate (2.7%), followed by 5 January (2.6%) and 4 January (2.4%). 5 January performs well on both page views and click-through rate. Changing N shows more or fewer days.

PROGRAM 3: Stacked Area Chart of User Interactions

Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

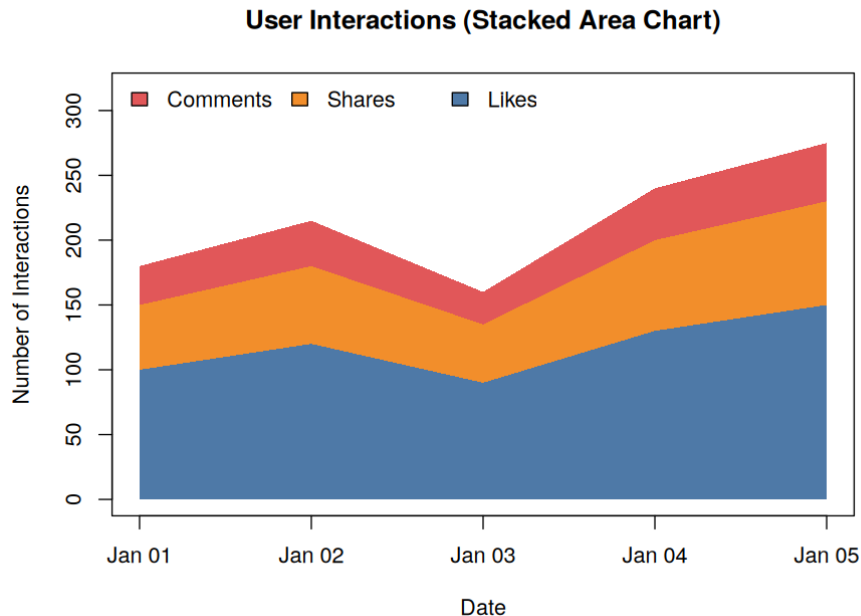
```
# Interaction data
likes    <- c(100, 120, 90, 130, 150)
shares   <- c(50, 60, 45, 70, 80)
comments <- c(30, 35, 25, 40, 45)

# Cumulative layers for stacking
x <- as.numeric(date)
c1 <- likes
c2 <- likes + shares
c3 <- likes + shares + comments

# Empty plot area
plot(date, c3, type = "n",
     ylim = c(0, max(c3) * 1.15),
     xlab = "Date",
     ylab = "Number of Interactions",
     main = "User Interactions (Stacked Area Chart)")

# Stacked areas
polygon(c(x, rev(x)), c(c3, rev(c2)), col = "#E15759", border = NA)
polygon(c(x, rev(x)), c(c2, rev(c1)), col = "#F28E2B", border = NA)
polygon(c(x, rev(x)), c(c1, rep(0, 5)), col = "#4E79A7", border = NA)

legend("topleft", legend = c("Comments", "Shares", "Likes"),
     fill = c("#E15759", "#F28E2B", "#4E79A7"),
     bty = "n", horiz = TRUE)
```



Output 3: User Interactions (Stacked Area Chart)

Observation: Total interactions follow the page-view pattern: they dip on 3 January (160) and peak on 5 January (275). Likes are the largest share every day (about 55%), followed by shares and comments.

Note: The dataset has no likes, shares or comments columns, so example interaction values were assumed.

PROGRAM 4: Interactive Website Traffic Dashboard

Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

```
library(shiny)

# Dataset
traffic <- data.frame(
  Date      = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                        "2023-01-04", "2023-01-05")),
  PageViews = c(1500, 1600, 1400, 1650, 1800),
  CTR       = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

ui <- fluidPage(
  titlePanel("Website Traffic Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      dateRangeInput("range", "Select Date Range:",
        start = min(traffic$Date), end = max(traffic$Date),
        min = min(traffic$Date), max = max(traffic$Date)),
      numericInput("n", "Top N Days (CTR):", value = 3, min = 1, max = 5),
      hr(),
      strong("Summary"),
      verbatimTextOutput("summary")
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("viewsPlot", height = "350px")),
        column(6, plotOutput("ctrPlot", height = "350px"))
      )
    )
  )
)

server <- function(input, output) {

  filtered <- reactive({
    traffic[traffic$Date >= input$range[1] & traffic$Date <= input$range[2], ]
  })

  output$viewsPlot <- renderPlot({
    d <- filtered()
    req(nrow(d) > 0)
    plot(d$Date, d$PageViews, type = "o", pch = 19, lwd = 2, col = "blue",
      xlab = "Date", ylab = "Page Views", main = "Daily Page Views")
    grid()
  }, width = 420, height = 340)

  output$ctrPlot <- renderPlot({
    d <- filtered()
    req(nrow(d) > 0)
    n <- min(input$n, nrow(d))
    top <- d[order(d$CTR, decreasing = TRUE)[1:n], ]
    barplot(top$CTR, names.arg = format(top$Date, "%b %d"), col = "darkorange",
      ylim = c(0, 3), xlab = "Date", ylab = "Click-through Rate (%)",
      main = paste("Top", n, "Days by CTR"))
  }, width = 420, height = 340)

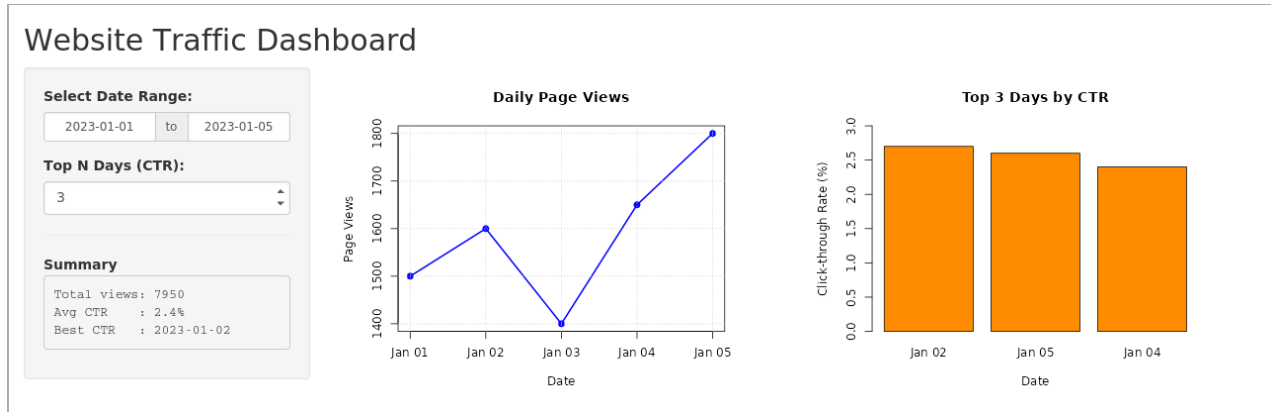
  output$summary <- renderText({
    d <- filtered()
    req(nrow(d) > 0)
    paste0("Total views: ", sum(d$PageViews), "\n",
```

```

    "Avg CTR    : ", round(mean(d$CTR), 2), "%\n",
    "Best CTR   : ", format(d$Date[which.max(d$CTR)])
  })
}

shinyApp(ui = ui, server = server)

```



Output 4: Interactive Website Traffic Dashboard (R Shiny)

Observation: The dashboard shows daily page views and the top N click-through-rate days together. Changing the date range or N updates both charts and the summary — for the full range, total page views are 7,950, average CTR is 2.4% and the best CTR day is 2 January.

Note: The question specifies Tableau; the equivalent dashboard was built with R Shiny. In Tableau: connect to the traffic data, create a line chart sheet (Date vs Page Views) and a bar chart sheet (Date vs Click-through Rate, sorted descending with a Top N filter), place both on a dashboard, and add a Date range filter applied to both sheets.

Result

The website traffic data was successfully visualized using a line chart, bar chart and stacked area chart, and an interactive dashboard combining the line and bar charts was created.